DATA SCIENCE CASE · NOVEMBER 2026

Fraud Prevention Case

Classifying transactions with Machine Learning

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ONE NUMBER TO START

At 10 pm, almost 1 in 3 transactions is a fraud.

And the score used today can’t tell the difference.

Source: 2025 transaction history (523,525 transactions)

THE CHALLENGE

Manual review can’t keep up

The review of suspicious transactions is mostly manual and does not scale. This creates two risks:

Frauds not caught in time

Financial loss for the company.

Good customers blocked

Bad experience and customer service cost.

Goal: a Machine Learning model that classifies transactions as fraud or not fraud.

THE CHALLENGE

Fraud is rare, and that makes it hard to find

523,525

transactions in 2025, with the fraud label

3.9%

of them are fraud

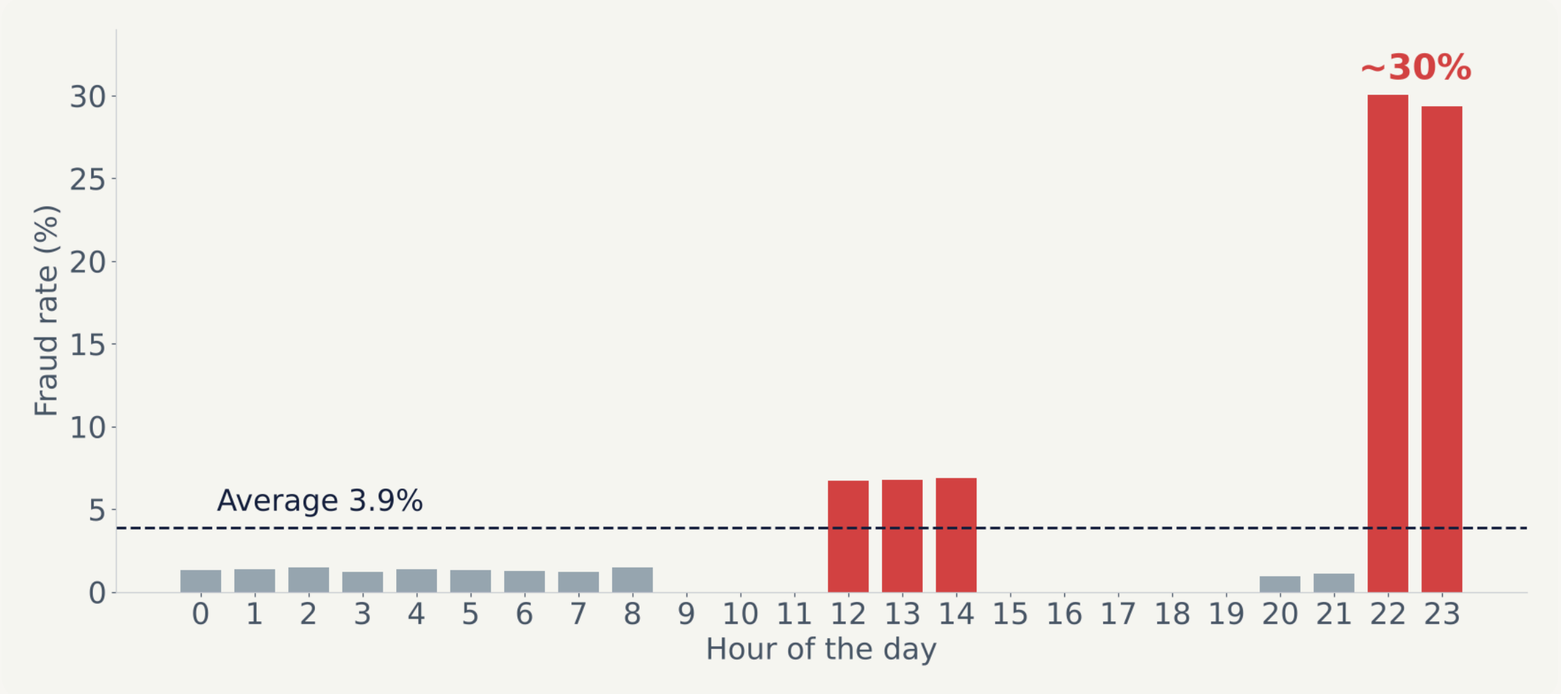
44,451

transactions in Jan/2026 to classify

Fraud is rare, so **accuracy is misleading**: a model that never flags fraud would be 96% “accurate” and useless. The model is evaluated by **precision** and **recall**.

WHAT THE DATA SHOWED

Fraud has a favorite time: 10–11 pm



~30%

of transactions at 10–11 pm are fraud, more than 7× the average.

Smaller peak at 12–2 pm (~7%). No fraud at 9–11 am and 3–7 pm.

WHAT THE DATA SHOWED

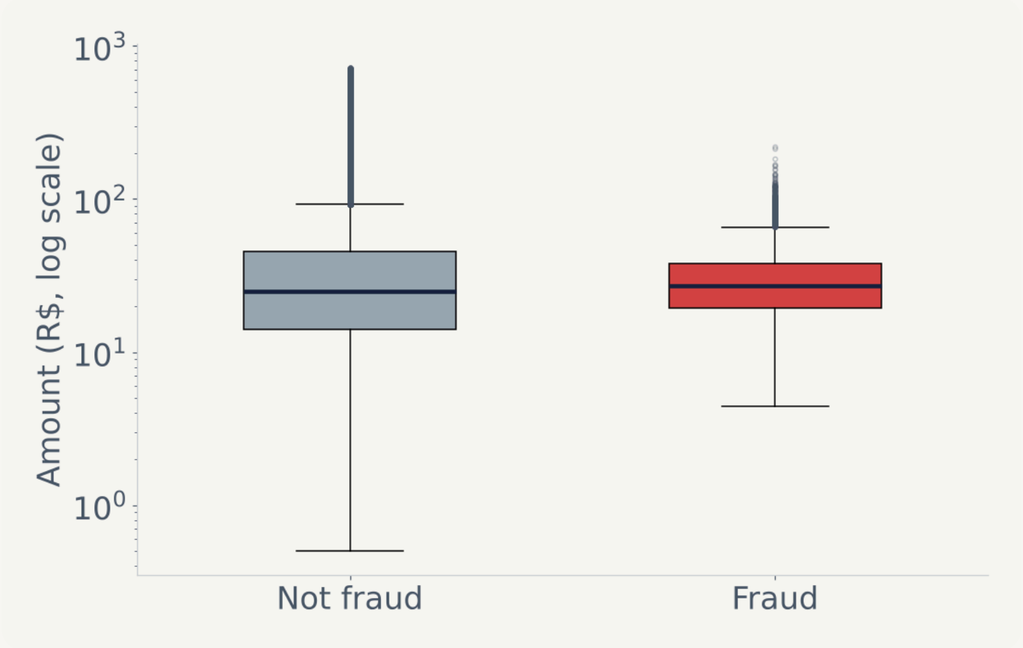
It also has a clear risk profile

Signal	Higher risk	Lower risk	Difference
Device	New device: 12.9%	Known device: 2.1%	6×
Shift	Night: 10.2%	Morning: 0.7%	15×
Merchant	Gas station: 7.5%	Pharmacy: 2.3%	3×
Region	Southeast: 4.9%	Northeast: 2.6%	2×
Device type	Mobile: 4.9%	Tablet: 2.1%	2×

Fraud rate by group in 2025 (average: 3.9%). Food & beverage is also high: 6.6%.

WHAT THE DATA SHOWED

And it tries to look like a normal purchase



No fraud above R$ 219, while legitimate purchases go up to R$ 723.

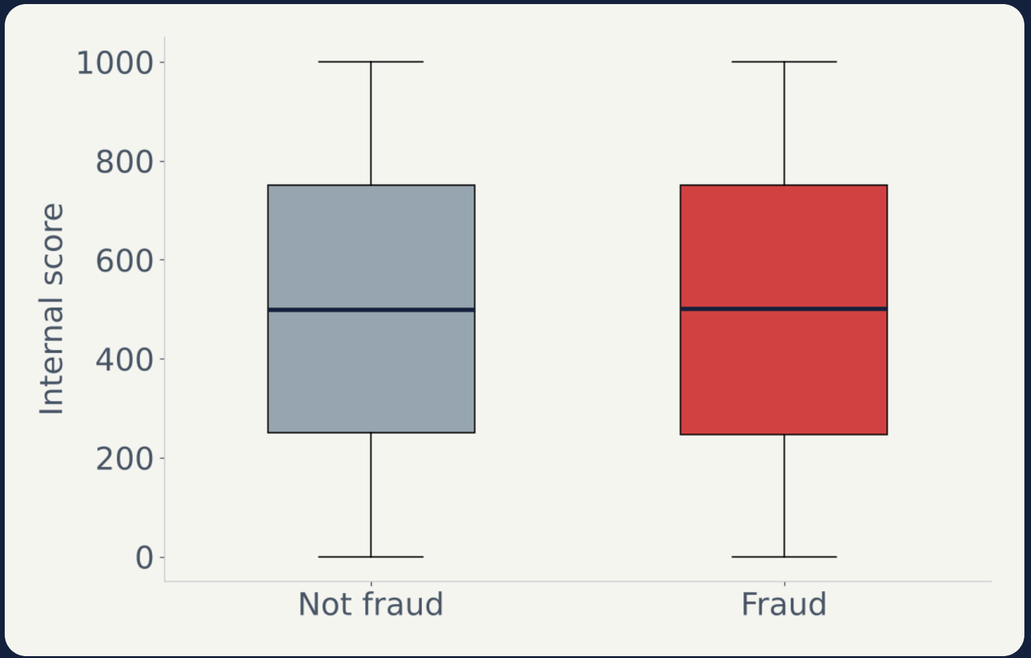
Close to usual spending: fraud is highest when the amount is near the customer’s own average.

December: 5.4% fraud, 44% above a typical month.

Amounts assumed in Brazilian reais (R$); the dataset does not state the currency.

THE BLIND SPOT

But today’s score can’t see any of this



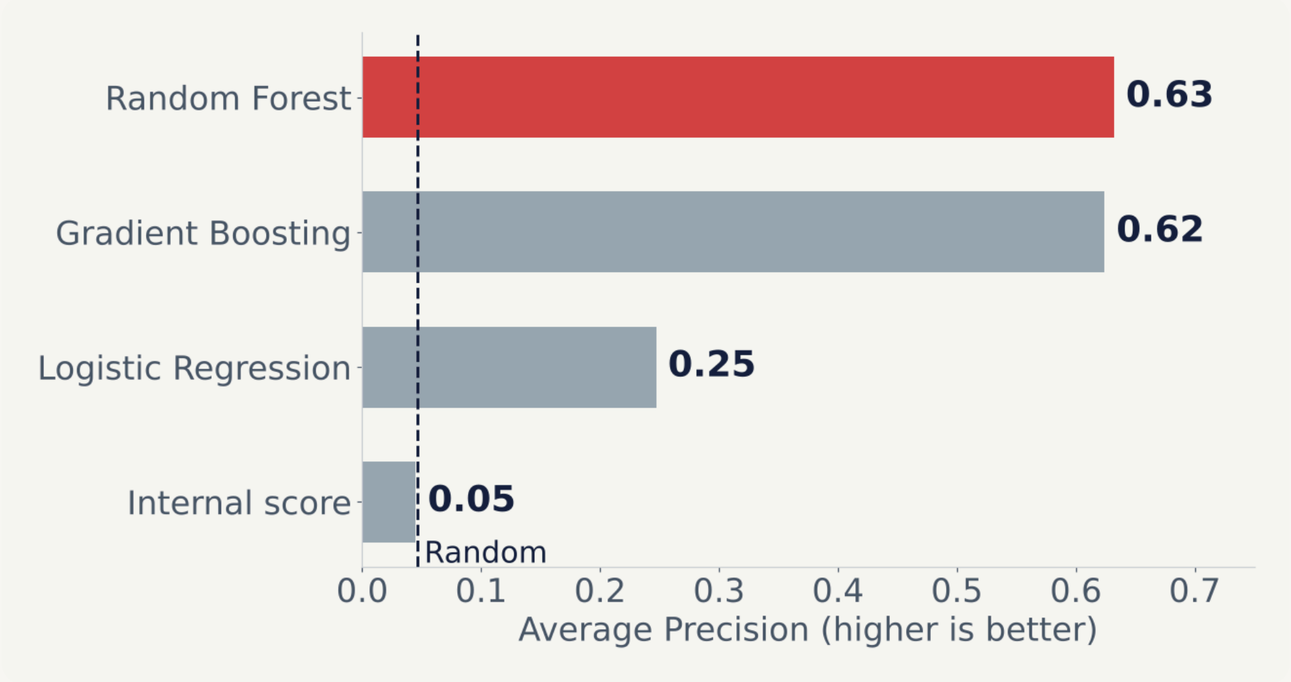
Fraud and non-fraud have the **same score distribution** (median 501 vs. 500).

Average Precision **0.05** and AUC **0.50**, the same as random.

Its method is unknown: worth checking with the team what it measures.

THE SOLUTION

A model that learns these patterns



Trained on Jan–Oct/2025 and tested on **Nov–Dec/2025**, data the model never saw.

Random Forest had the best result (0.63, about 14× random) and is simpler to explain than Gradient Boosting.

New features: hour of the day and each customer's spending history.

THE IMPACT · THRESHOLD 0.2

3 of every 4 frauds caught, reviewing only 7%

77%

of frauds detected (3,094 of 3,991)

~50%

of the alerts are real fraud

7.1%

of transactions get an alert

January/2026: 3,431 transactions flagged (7.7%), close to the test period.

WHAT TO DO

Let the team focus on the doubtful cases

Use the model with **risk ranges**, so the team only reviews the doubtful cases:

HIGH PROBABILITY

Automatic block

MEDIUM PROBABILITY

Manual review

LOW PROBABILITY

Automatic approval

- More monitoring at 10–11 pm and for new devices.
- Review with the team how the internal score is calculated.

WHAT COMES NEXT

Limitations and next steps

Limitations

- The threshold was chosen on the test set itself.
- The fraud label was assumed to be reliable.
- The hourly pattern is very sharp: confirm it with new data.

Next steps

- Retrain the model every month with recent data.
- Set the threshold using the real cost of each error.
- Monitor the model performance month by month.
- Add new data sources (device, location): the model variations I tested gave no gain (notebook, section 7.7).

Thank you!

Questions?

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